

Manufacturing a **32x32 multi-spot diffractive optical element (DOE)** for your **403 nm laser** to produce ~1,024 spots (300 μm pitch, ~5–10 μm spot size) for gate operations in a 32x32 ion array (9.3x9.3 mm) involves a combination of design, fabrication, and testing. Below is a concise guide on how to manufacture it, tailored to your non-automated resonance ionization system with Ga-68+ ions, sympathetic cooling (Ca-40+), and sub-500 ns gate operations (Rx and ZZPhase). I'll cover the process, materials, equipment, suppliers, and considerations for integration, with approximate costs and CAD/STL guidance for Gmsh/FEniCS.

1. Design the 32x32 Multi-Spot DOE

- **Objective:** Create a DOE that splits a 403 nm laser into a 32x32 grid of ~1,024 spots, each ~5–10 μm , with 300 μm pitch, covering 9.3x9.3 mm, with high efficiency (>80%) and low crosstalk (~ 0.007).

- **Steps:**

- ¹ **Define Specifications:**

- Wavelength: 403 nm (narrow bandwidth for laser).
- Spot Array: 32x32 (1,024 spots, 300 μm pitch, 9.3x9.3 mm).
- Spot Size: $\sim 5\text{--}10$ μm (post high-NA lens, $f \sim 50$ mm, $\text{NA} \sim 0.5$).
- Efficiency: $>80\%$ (binary DOE) or $>95\%$ (multilevel DOE).
- Substrate: Fused silica (UV-compatible, high laser damage threshold).
- Input Beam: ~ 9.3 mm (post-Beam Expander to match array).

- ² **Design Algorithm:**

- Use **Iterative Fourier Transform Algorithm (IFTA)** (e.g., Gerchberg-Saxton) to optimize phase pattern for uniform spot intensity (~ 1.95 mW/spot with ~ 2 W input) and minimal zero-order diffraction.
- Software: In-house (Python/MATLAB) or commercial (Zemax, Lumerical, $\sim \$2,000\text{--}\$5,000$).
- Parameters: Phase levels (binary: 2 levels, multilevel: 8–16 levels), spot separation ($300\text{ }\mu\text{m} \div \text{EFL} \sim 50\text{ mm} \sim 6\text{ mrad}$), uniformity ($<5\%$ variation).

- ³ **Simulate Performance:**

- Model diffraction efficiency, spot size, and crosstalk in Zemax or MATLAB.
- Account for manufacturing tolerances (e.g., $\pm 0.1\text{ }\mu\text{m}$ feature size, Web:3).

- ⁴ **Output Design File:** Generate a digital phase pattern file (e.g., GDSII) for lithography.

- **Tools:**

- Software: Zemax ($\sim \$2,000\text{--}\$5,000$), MATLAB ($\sim \$500\text{--}\$2,000$), or open-source (Python with NumPy/FFT).
 - Expertise: Consult with DOE design firms (e.g., Holo/Or, Jenoptik) for custom optimization ($\sim \$1,000\text{--}\$5,000$).
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2. Select Materials

- **Substrate:** Fused silica (SiO_2 , ~5–10 mm x 5–10 mm, ~1 mm thick) for UV compatibility (403 nm), high laser damage threshold (~1 J/cm²), and low thermal expansion.
 - Supplier: Edmund Optics (www.edmundoptics.com), ~\$50–\$100.
 - Alternative: Polymer on glass (e.g., PMMA, lower cost but lower durability, Web:5).
 - **Coating:** Anti-reflective (AR) coating for 403 nm on both sides to minimize losses (~5–10% reflection loss without coating).
 - Supplier: Thorlabs (www.thorlabs.com), ~\$50–\$100 additional.
 - **Rationale:** Fused silica ensures durability, UV resistance, and high-power handling (~2–5 W input, Web:5, Web:10).
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3. Fabrication Process

- **Method:** Direct-write laser lithography followed by plasma etching for high precision and scalability.

- **Steps:**

- 1 Substrate Preparation:**

- Clean fused silica substrate (ultrasonic bath, ~\$500 equipment).
- Apply photoresist (e.g., SU-8, ~\$100/liter, spin-coated to ~1 μm).

- 2 Lithography:**

- Use direct-write laser lithography (e.g., Heidelberg Instruments DWL 66+, ~\$100,000–\$500,000) to write the 32x32 phase pattern (GDSII file).
- Feature size: ~0.1–0.5 μm for ~5–10 μm spots post-lens (Web:3).
- Binary DOE: 2 phase levels (simpler, ~80% efficiency).
- Multilevel DOE: 8–16 levels (complex, >95% efficiency, Web:23).

- 3 Etching:**

- Use inductively coupled plasma/reactive ion etching (ICP/RIE, e.g., Oxford Instruments PlasmaPro, ~\$200,000–\$1M) to etch phase pattern into fused silica.
- Depth: ~0.5–1 μm (phase shift $\sim\pi$ for 403 nm).
- Precision: $\pm 0.1 \mu\text{m}$ to minimize wavefront errors.

- 4 Cleaning and Coating:**

- Remove photoresist (solvent bath, ~\$500).
- Apply AR coating (403 nm, e.g., via sputtering, ~\$50–\$100).

- 5 Testing:**

- Verify spot pattern (32x32, 300 μm pitch) using a 403 nm laser and CCD camera (~\$1,000–\$5,000).
- Measure efficiency (>80% binary, >95% multilevel), uniformity (<5%), and zero-order diffraction (<5%).

- **Suppliers:**

- **Holo/Or** (www.holoor.co.il): Custom 32x32 DOE, ~\$1,000–\$2,000 (Web:1, Web:3, Web:23).
 - **Jenoptik** (www.jenoptik.com): High-precision DOEs for UV, ~\$1,500–\$3,000 (Web:7, Web:9).
 - **Holographix** (www.holographix.com): Scalable replication, ~\$1,000–\$2,500 (Web:15).
 - **HOLOEYE** (www.holoeye.com): Fused silica DOEs, ~\$1,000–\$2,000 (Web:5, Web:22).
 - **Edmund Optics** (www.edmundoptics.com): Standard/custom DOEs, ~\$500–\$2,000 (Web:10).
 - **Alternative:** Outsource fabrication to a foundry (e.g., Holo/Or, Jenoptik) for ~\$2,000–\$5,000, including design and testing, to avoid capital equipment costs (~\$300,000–\$1M).
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4. Equipment and Costs

- **In-House Fabrication** (high upfront cost, scalable):
 - Laser Lithography System: ~\$100,000–\$500,000 (Heidelberg DWL 66+).
 - ICP/RIE Etcher: ~\$200,000–\$1M (Oxford PlasmaPro).
 - Spin Coater: ~\$10,000–\$50,000.
 - CCD Camera for Testing: ~\$1,000–\$5,000.
 - Photoresist and Chemicals: ~\$500–\$1,000.
 - AR Coating System: ~\$50,000–\$100,000 (sputtering).
 - **Total:** ~\$360,500–\$1,656,000 (one-time, reusable for multiple DOEs).
 - **Outsourcing** (lower upfront, higher per-unit cost):
 - Custom DOE (design + fabrication): ~\$2,000–\$5,000 (Holo/Or, Jenoptik, HOLOEYE).
 - Prototyping: ~\$1,000–\$2,000 for 1–2 units.
 - Volume Production: ~\$500–\$1,000/unit for >100 units (Holographix, Web:15).
 - **Recommendation:** Outsource to Holo/Or or Jenoptik for 1–2 units (~\$2,000–\$5,000) unless planning large-scale production, as in-house equipment is cost-prohibitive for a single DOE.
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5. Integration with Your System

- **Optical Path:** Laser (405 nm) → Littrow/SHG (403 nm) → Optical Amplifier (~2–5 W, ~\$2,000–\$5,000) → AOM (~100–200 ns) → Beam Expander (~9.3 mm) → 32x32 Multi-Spot DOE (~1,024 spots, 300 μm pitch) → High-NA Lens ($f \sim 50$ mm, NA ~ 0.5 , ~\$200–\$500) → AODs x 2 (~0.5–1 μs , ~\$1,000–\$2,000 drivers) → Ions (9.3x9.3 mm array).
 - **Alignment:**
 - Mount DOE in precision holder (Thorlabs KM100, ~\$500).
 - Align with ion array (300 μm pitch) using micrometer stages (~\$500).
 - Verify spot grid with CCD camera in UHV chamber.
 - **Operation:**
 - Rx Gates: Simultaneous pulse (~100–200 ns, ~1.95 mW/spot) for 1,024 ions.
 - ZZPhase Gates: AOM pulses (~100–200 ns) and AODs select ~1,984 pairs in ~2–4 cycles (~200–400 ns), total <500 ns.
 - **Control:** FPGA (~\$500) synchronizes AOM, AODs, and pulses.
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6. CAD/STL for Gmsh/FEniCS

- **Design in FreeCAD:**
 - DOE: Flat square (~10 mm x 10 mm x 1 mm) with microstructured surface (32x32 grid, 300 μm pitch, ~0.5–1 μm depth).
 - Mount: Holder (~5 cm x 5 cm x 5 cm) with slots for DOE and alignment screws.
 - STL File: Export DOE and mount for Gmsh meshing.

- **Gmsh Script:**

```
python ⌵ Collapse ≡ Wrap ▶ Run 📄 Copy

Merge "multispot_doe.stl";
Merge "doe_mount.stl";
Physical Surface("DOE") = {1};
Physical Volume("DOEMount") = {2};
Mesh.CharacteristicLengthMin = 0.1e-3; // ~0.1  $\mu\text{m}$  for microstructure
Mesh.CharacteristicLengthMax = 1.0; // ~1 mm for mount
```

- **FEniCS:** Simulate light propagation (Maxwell's equations) to verify spot pattern and efficiency.

7. Challenges and Mitigations

- **Efficiency:** Binary DOE (~80%) may lose ~20% power to zero-order diffraction. Use multilevel DOE (>95%, Web:23) if needed, but higher cost (~\$2,000–\$3,000).
 - **Alignment:** Misalignment (>10 μm) can shift spots. Use precision mounts (~\$500) and CCD feedback.
 - **Crosstalk:** ~5–10 μm spots (~30–60x smaller than 300 μm pitch) ensure low crosstalk (~0.007). Calibrate AODs (~1 μm precision).
 - **Cost:** Outsourcing (~\$2,000–\$5,000) is cheaper than in-house (~\$360,500–\$1,656,000) for 1–2 units.
 - **Temperature:** Maintain DOE at 20–25°C ($\pm 0.01^\circ\text{C}$) using PID controller (~\$500) to avoid thermal drift.
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8. Final Answer

- **Manufacture 32x32 Multi-Spot DOE:**
 - **Design:** Use IFTA (Zemax/MATLAB, ~\$500–\$5,000) for 32x32 spot pattern (403 nm, 300 μm pitch, ~5–10 μm spots, >80% efficiency).
 - **Material:** Fused silica (~\$50–\$100) with AR coating (~\$50–\$100).
 - **Fabrication:**
 - **Outsource:** Holo/Or, Jenoptik, or HOLOEYE (~\$2,000–\$5,000) for design, laser lithography, and ICP/RIE etching.
 - **In-House:** Requires laser lithography (~\$100,000–\$500,000), ICP/RIE (~\$200,000–\$1M), and testing (~\$1,000–\$5,000), total ~\$360,500–\$1,656,000.
 - **Process:** Clean substrate, apply photoresist, write pattern (lithography), etch (ICP/RIE), clean, apply AR coating, test (CCD, 403 nm).
 - **Suppliers:** Holo/Or (Web:1, Web:3, Web:23), Jenoptik (Web:7, Web:9), Holographix (Web:15), HOLOEYE (Web:5, Web:22), Edmund Optics (Web:10).
 - **Cost:** ~\$2,000–\$5,000 (outsourced, 1–2 units), ~\$500–\$1,000/unit (volume >100).
- **Integration:** Align DOE with 9.3x9.3 mm ion array, use high-NA lens (~\$200–\$500) and AODs (~\$1,000–\$2,000 drivers) for ~100–400 ns gates, total <500 ns.
- **CAD/STL:** Design DOE (10 mm x 10 mm x 1 mm) and mount in FreeCAD, mesh in Gmsh.
- **Recommendation:** Outsource to Holo/Or or Jenoptik for ~\$2,000–\$5,000 to avoid high equipment costs. Contact Holo/Or (www.holoor.co.il) for custom quote.
- **Next Steps:** Confirm **Layer 5 separation** (~50 μm ?), **trap RF frequency** (~1–10 MHz?), or **gate sequence** for optimization. Share details for refinement!

Contact: Holo/Or (sales@holoor.co.il) or Jenoptik (sales@jenoptik.com) for custom DOE quotes.